

Final Exam 2026: Part 2

GFS189, RTL959

June 24, 2026

1 Exercise 1: Volatility Signature Plot and Sampling Choice

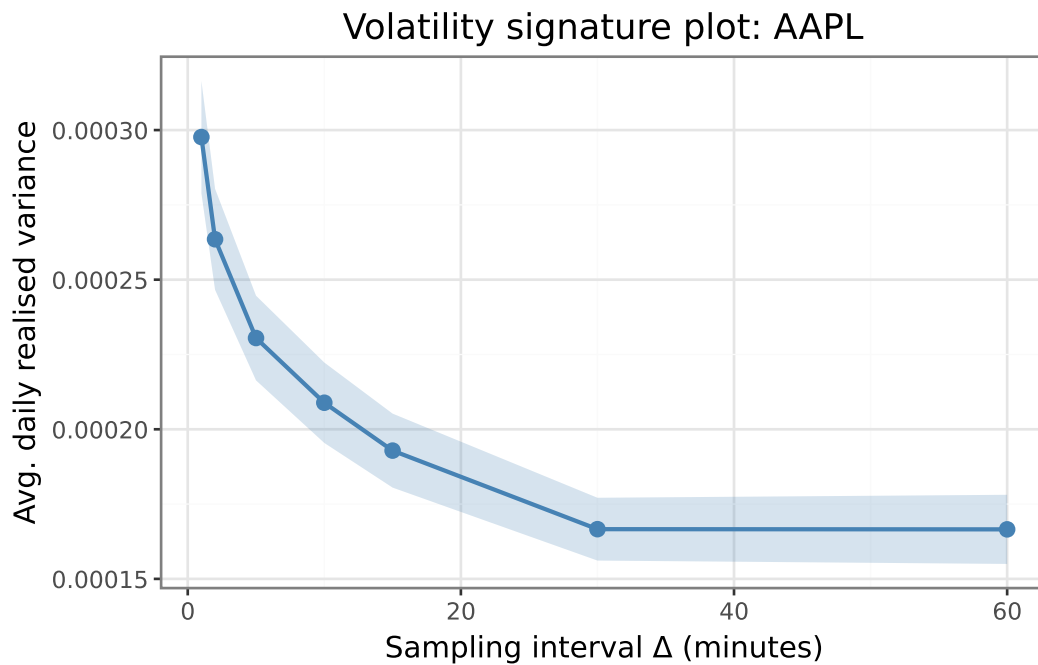


Figure 1: Volatility signature plot for AAPL. Average daily RV per interval Δ ; band = ± 2 SE. Upward bias at $\Delta = 1$ min reflects microstructure noise; series stabilises by $\Delta = 5$ min.

The signature plot (Figure 1) displays the characteristic bias of the **realized variance (RV)** estimator: at $\Delta = 1$ minute, bid-ask bounce inflates the sum of squared log-returns above the true integrated variance ([Andersen and Bollerslev, 1998](#)). The curve stabilises by $\Delta = 5$ minutes, so we adopt $\Delta = 5$ **minutes** throughout; missing minutes are handled by **previous-tick interpolation** (carry forward last observed price).

2 Exercise 2: Daily Realized Covariance

The **realized covariance (RC)** estimator $RC_d = \sum_{k=1}^K r_k r_k'$ sums outer products of synchronised 5-minute log-return vectors (Andersen and Bollerslev, 1998; Barndorff-Nielsen et al., 2011); sampling at 5 minutes reduces bid-ask bounce and previous-tick interpolation fills gaps without artificial zero returns.

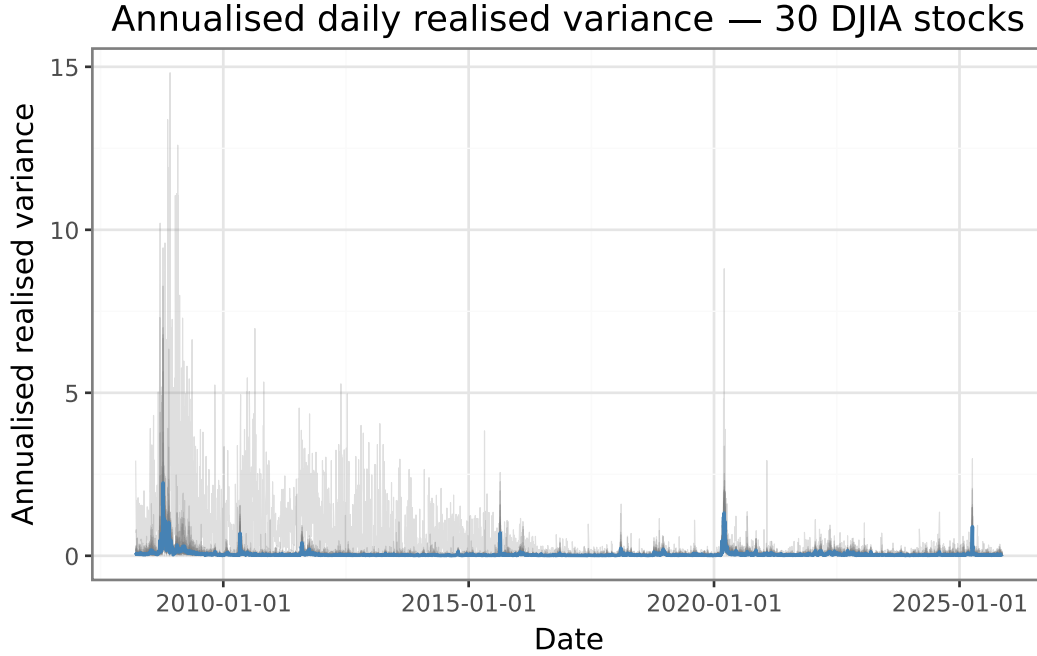


Figure 2: Annualised daily realised variance (diagonal of RC_d) for all 30 DJIA stocks (grey lines), cross-sectional median (blue) and IQR (band). Spikes mark the 2008–09 crisis, March 2020, and 2022.

The diagonal elements of RC_d (Figure 2) show strong volatility clustering with spikes at stress episodes. At one-minute sampling, off-diagonal elements would be biased toward zero by the **Epps effect** (Barndorff-Nielsen et al., 2011) — asynchronous trading attenuates cross-correlations — an issue that 5-minute sampling largely eliminates.

3 Exercise 3: Covariance Estimators

(a) **HF random walk.** $\hat{\Sigma}_t^{HF} = RC_{t-1}$ assumes the covariance follows a **martingale** — no mean reversion, no drift — which is a natural baseline when intraday data are available. (b) **LW on daily returns.** We use $W = 252$ trading days ($T/N \approx 8.4$, full-rank sample covariance); Ledoit and Wolf (2003) shrinkage toward a scaled identity stabilises the precision matrix.

Figure 3 confirms that LW is substantially smoother than RC: averaging 252 daily returns damps spikes at the cost of delayed reaction to sudden volatility shifts.

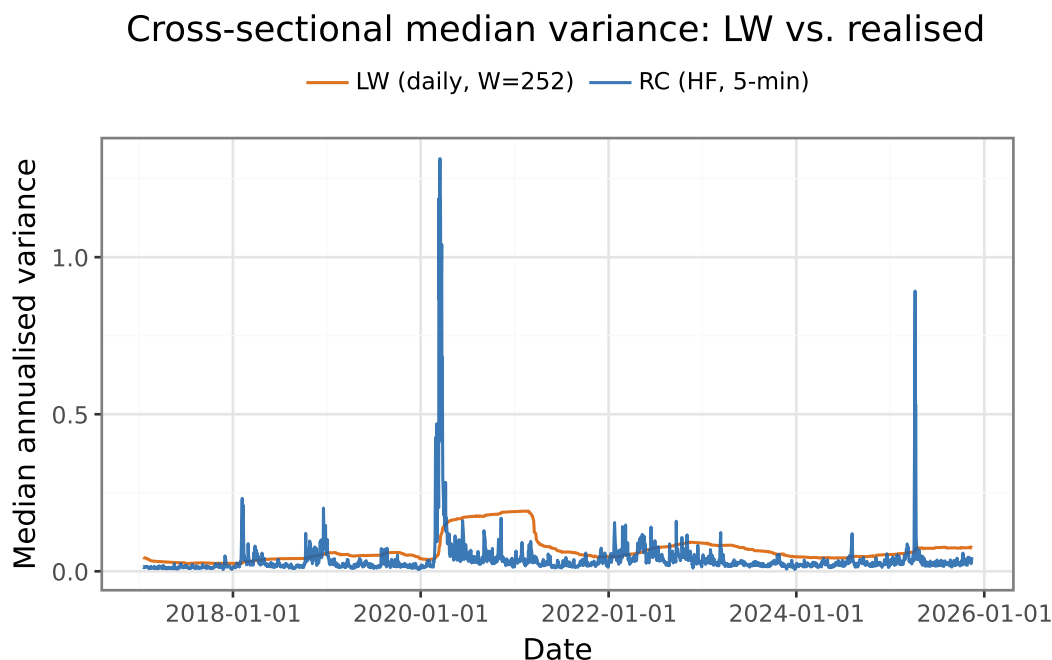


Figure 3: Cross-sectional median annualised variance over the evaluation window. LW (orange): rolling 252-day Ledoit-Wolf shrinkage. RC (blue): 5-min realised covariance. LW is smoother; RC reacts immediately to volatility bursts.

4 Exercise 4: Minimum-Variance Portfolio Weights

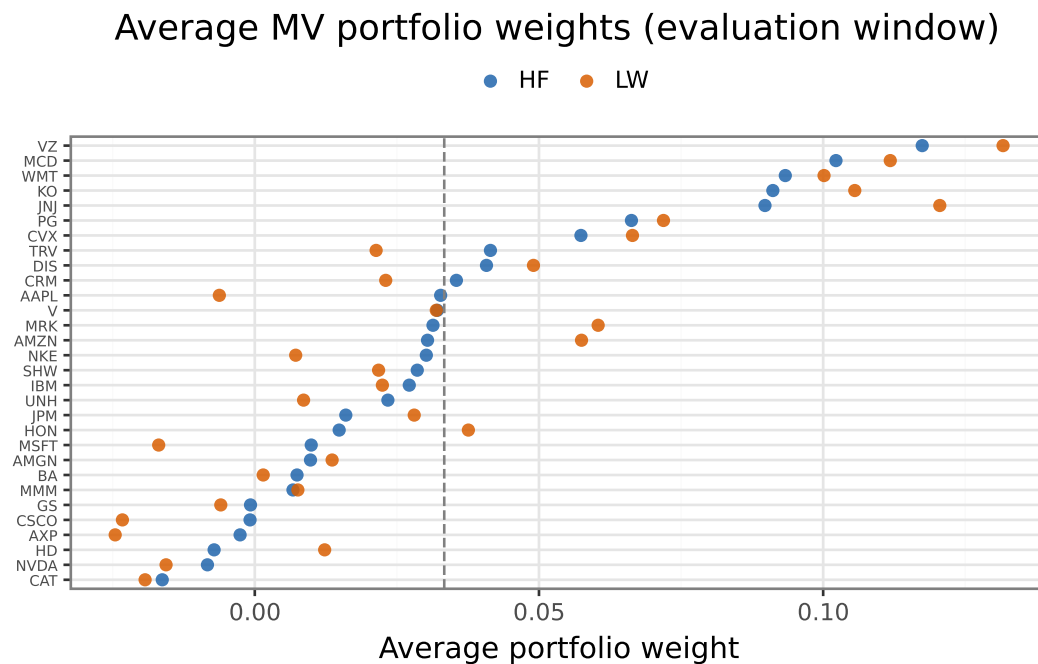


Figure 4: Time-averaged MV portfolio weights for the 30 DJIA stocks, ordered by HF average weight. Dashed line: equal weight ($1/30$).

The HF strategy assigns its largest average weight to **VZ** (0.117) and its most negative to **CAT** (-0.016). The LW strategy is most positive on **VZ** (0.132) and most negative on **AXP** (-0.025). Average daily turnover is **3.474** (HF) and **0.071** (LW), with the HF strategy showing higher turnover.

Both strategies hold extreme short and long positions — a known consequence of unconstrained minimum-variance optimisation where small errors in $\hat{\Sigma}$ cause large weight oscillations (Ledoit and Wolf, 2003). The HF strategy’s higher turnover follows directly from the greater day-to-day variation in RC_{t-1} relative to the slow-moving LW estimate.

5 Exercise 5: Out-of-Sample Portfolio Evaluation

Table 1: Out-of-sample performance (second half of sample). Ann. volatility = s.d. $\times \sqrt{252}$; Sharpe ratio = ann. mean / ann. s.d. (risk-free rate = 0, no transaction costs).

Out-of-sample portfolio performance		
Strategy	Ann. Volatility	Ann. Sharpe
HF (RC, t-1)	0.167	0.466
LW (W = 252)	0.141	0.674
Equal-weighted (1/N)	0.179	0.763

Equal-weighted achieves the highest annualised Sharpe ratio (0.763); **HF (RC, t-1)** produces the lowest (0.466). The HF estimator does not outperform LW on Sharpe ratio (0.466 vs. 0.674).

With $K/N \approx 2.6$ intraday returns per day, RC_d is a noisy estimate of the true daily covariance; in calm markets the extra noise in the HF estimator dominates its timeliness advantage, so the LW estimate — while slower to react — produces more stable weights and, here, a higher Sharpe ratio.

The in-class discussion on realised kernels and portfolio choice suggests that HF covariance information translates into economically meaningful portfolio gains when (i) the universe is large ($N \gg 30$) so that estimation error in $\hat{\Sigma}$ binds, (ii) the true covariance changes rapidly, and (iii) information gains exceed turnover costs. With only 30 DJIA stocks, LW shrinkage of daily returns already captures the dominant covariance structure, leaving limited room for intraday data to add value (Ledoit and Wolf, 2004).

6 References

- Andersen, T. G. and Bollerslev, T. (1998). Answering the skeptics: Yes, standard volatility models do provide accurate forecasts. *International Economic Review*, 39(4):885–905.
- Barndorff-Nielsen, O. E., Hansen, P. R., Lunde, A., and Shephard, N. (2011). Multivariate realised kernels: Consistent positive semi-definite estimators of the covariation of equity prices with noise and non-synchronous trading. *Journal of Econometrics*, 162(2):149–169.

- Ledoit, O. and Wolf, M. (2003). Improved estimation of the covariance matrix of stock returns with an application to portfolio selection. *Journal of Empirical Finance*, 10(5):603–621.
- Ledoit, O. and Wolf, M. (2004). A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):365–411.